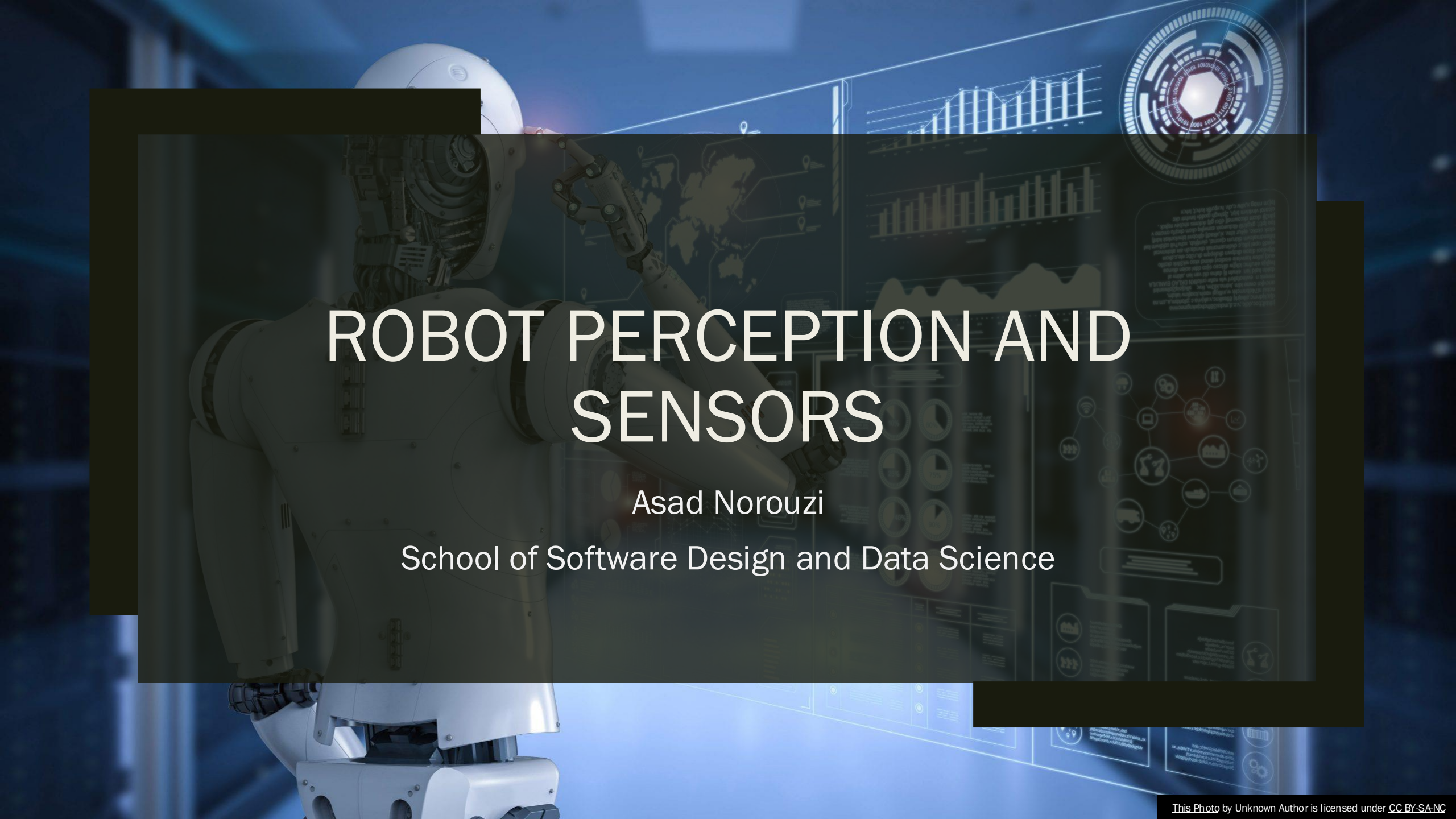


A white humanoid robot is shown from the waist up, positioned on the left side of the frame. It is interacting with a large, futuristic digital interface that occupies the right and center portions of the image. The interface displays various data visualizations, including bar charts, line graphs, a world map, and a circular radar-like chart in the top right corner. The background is a dark blue, slightly blurred, suggesting a high-tech environment. The overall aesthetic is futuristic and technological.

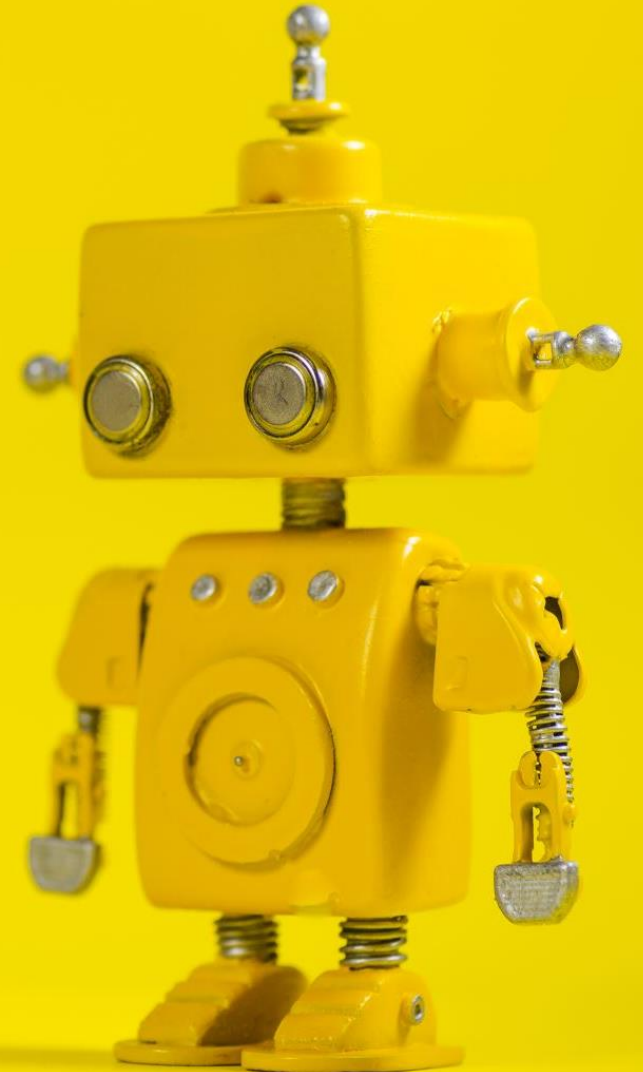
ROBOT PERCEPTION AND SENSORS

Asad Norouzi

School of Software Design and Data Science

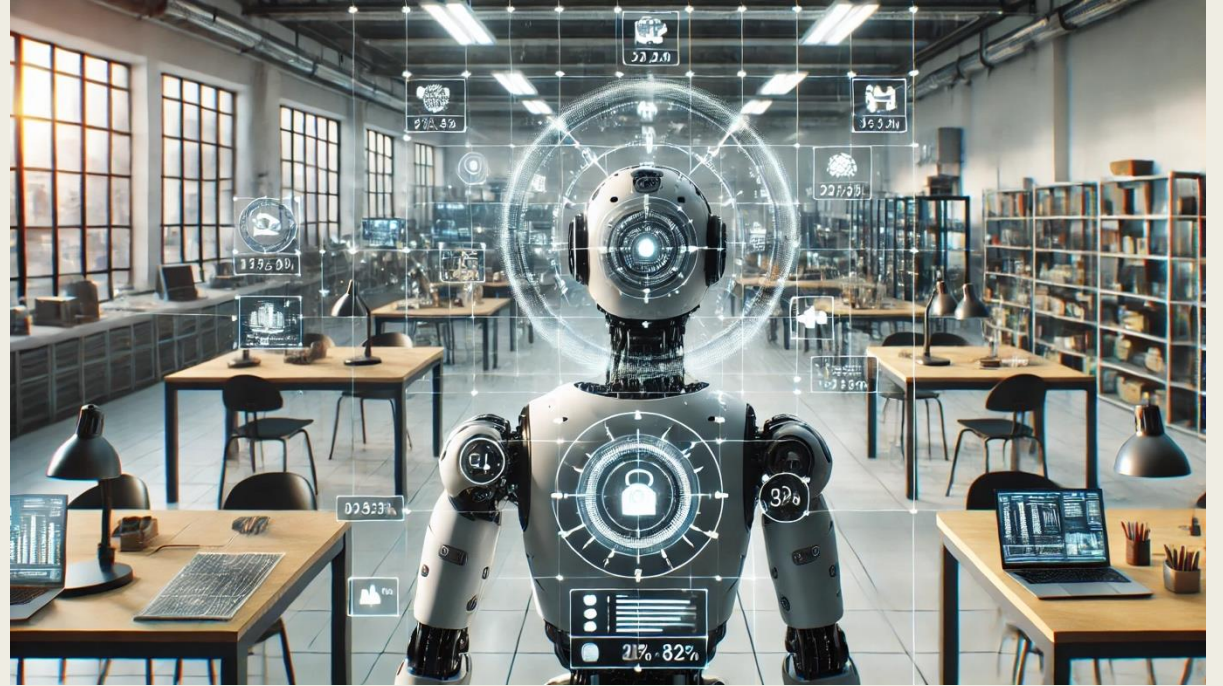
Agenda

- Introduction to Robot Perception
- Types of Sensors in Robotics
- Vision Sensors
- Proprioceptive Sensors
- Contact and Force Sensors
- Sensor Fusion
- Perception Systems
- Challenges in Robot Perception
- Summary
- Q&A



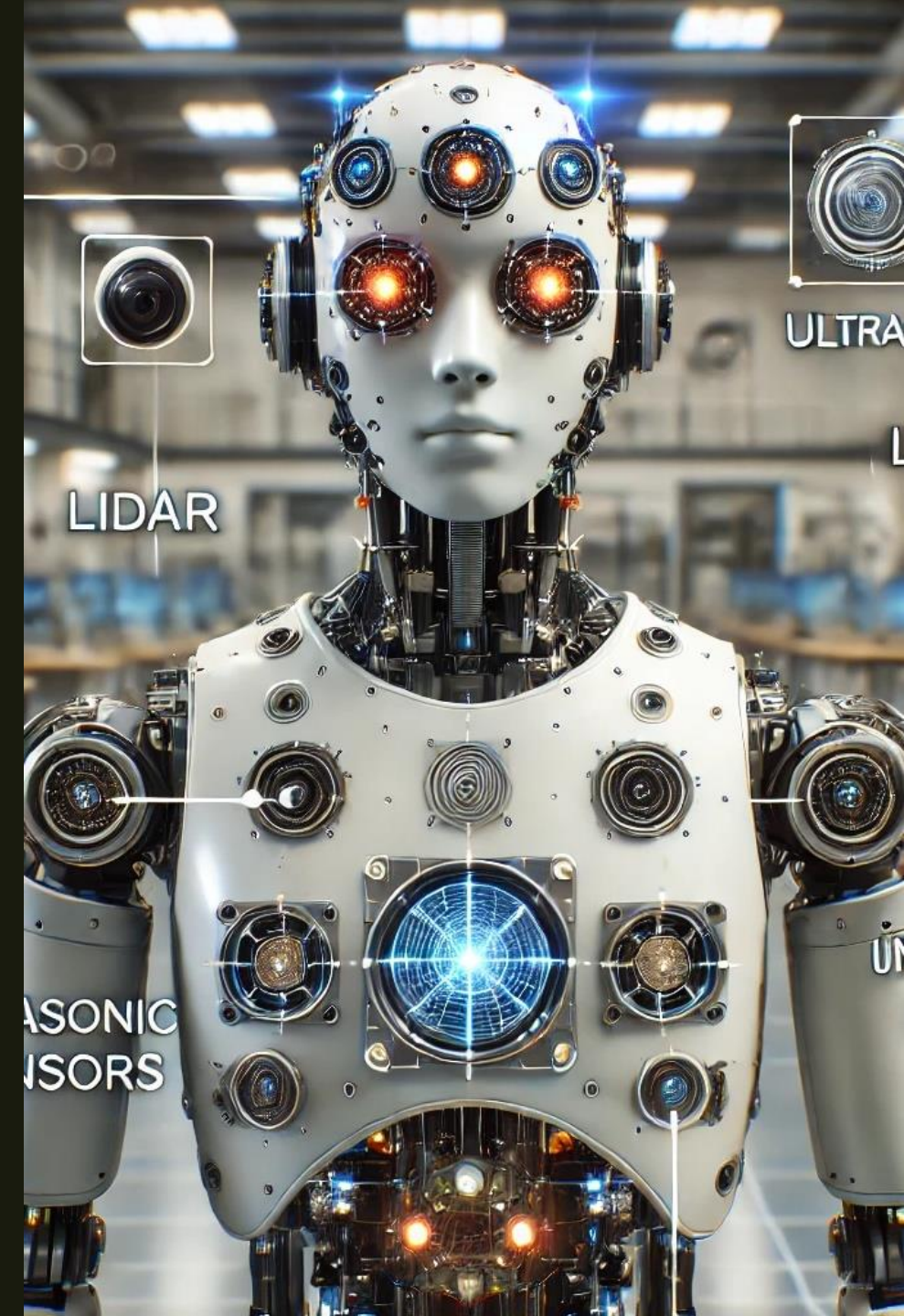
Introduction to Robot Perception

- Robot perception is the process by which robots interpret data from sensors to understand their environment and make informed decisions.
- Essential for autonomous operation, enabling robots to navigate, interact with objects, and perform tasks effectively.



Types of Sensors in Robotics

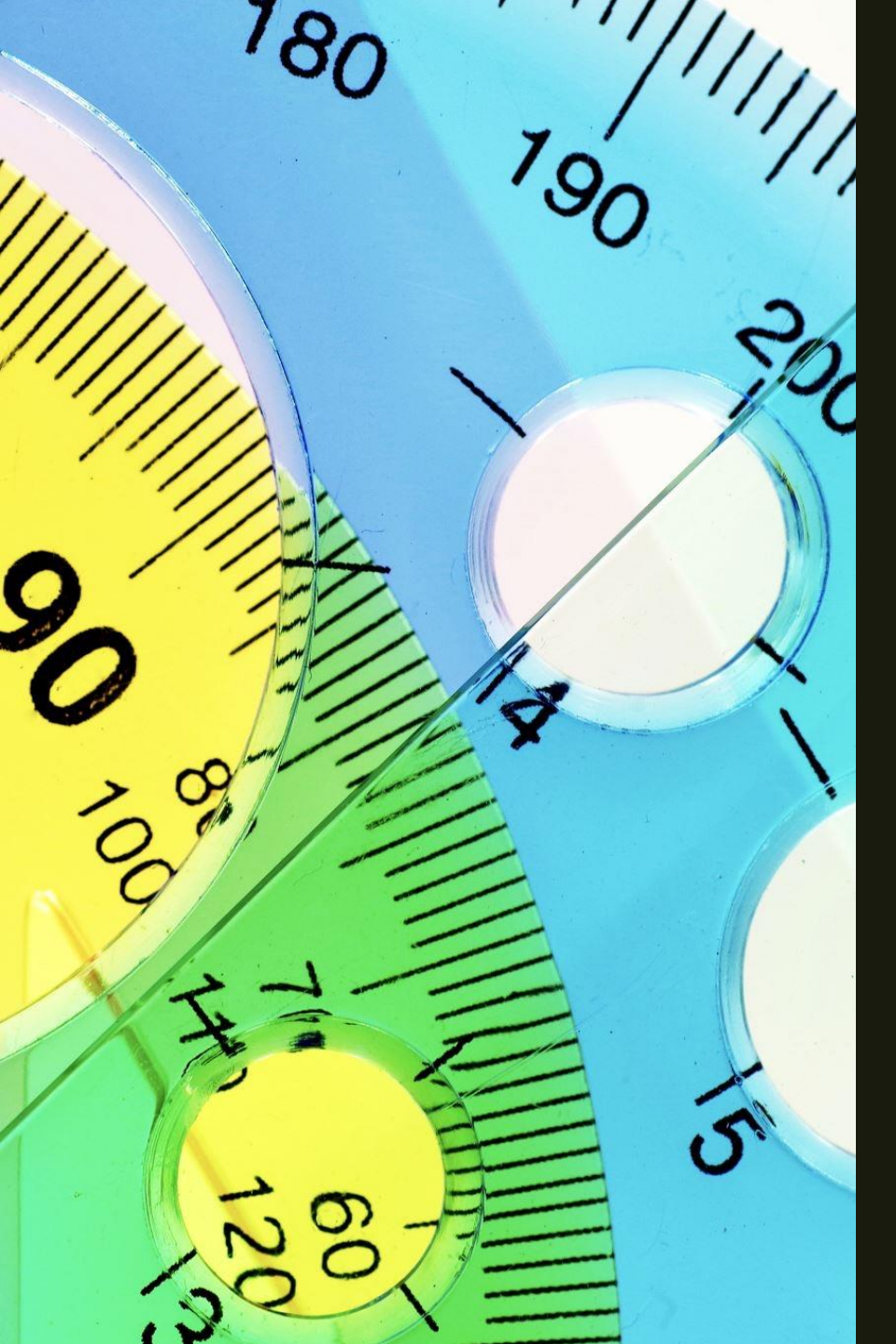
- **Proprioceptive Sensors:**
 - Measure internal states of the robot, such as joint angles, velocities, and accelerations.
 - Encoders, accelerometers, gyroscopes.
- **Exteroceptive Sensors:**
 - Measure external information from the environment.
 - Cameras, LiDAR, ultrasonic sensors, infrared sensors.
- **Contact Sensors:**
 - Detect physical contact or touch.
 - Tactile sensors, force sensors, bump sensors.
- **Environmental Sensors:**
 - Measure environmental conditions.
 - Temperature sensors, humidity sensors, gas sensors.



Vision Sensors

- **Cameras:**
 - *Capture images or videos of the environment.*
 - **Types:** *Monocular cameras, stereo cameras, RGB-D cameras.*
 - **Applications:** *Object recognition, navigation, mapping.*
- **LiDAR (Light Detection and Ranging):**
 - *Uses laser pulses to measure distances and create 3D maps.*
 - **Applications:** *Autonomous vehicles, drones, mapping.*
- **Infrared Sensors:**
 - *Detect heat signatures and measure distances.*
 - **Applications:** *Night vision, obstacle detection.*



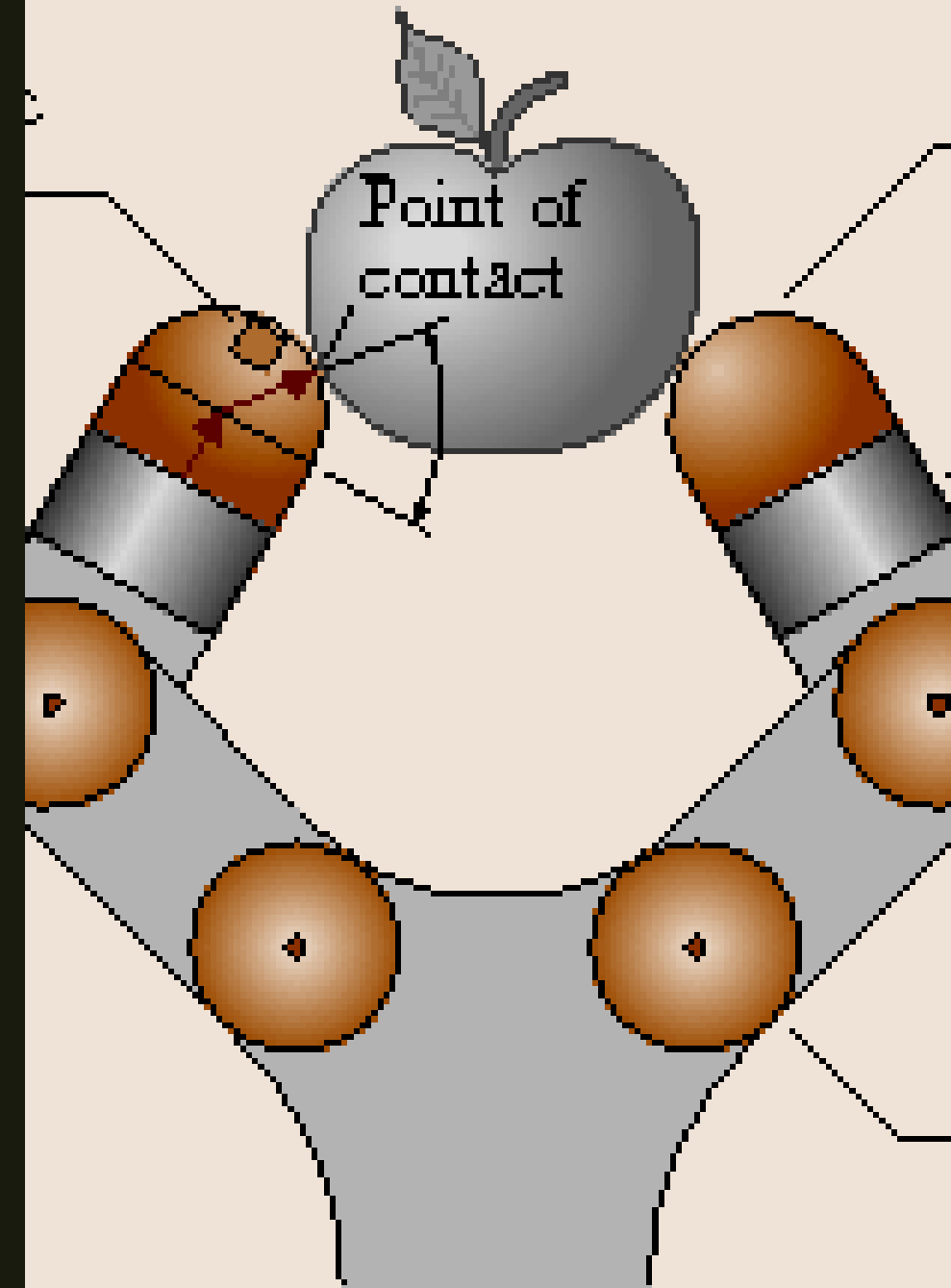


Proprioceptive Sensors

- **Encoders:**
 - *Measure the rotational position of a shaft.*
 - ***Types:** Incremental encoders, absolute encoders.*
 - ***Applications:** Joint angle measurement, wheel rotation tracking.*
- **Accelerometers:**
 - *Measure linear acceleration.*
 - ***Applications:** Motion detection, vibration monitoring.*
- **Gyroscopes:**
 - *Measure rotational velocity.*
 - ***Applications:** Orientation tracking, inertial navigation.*

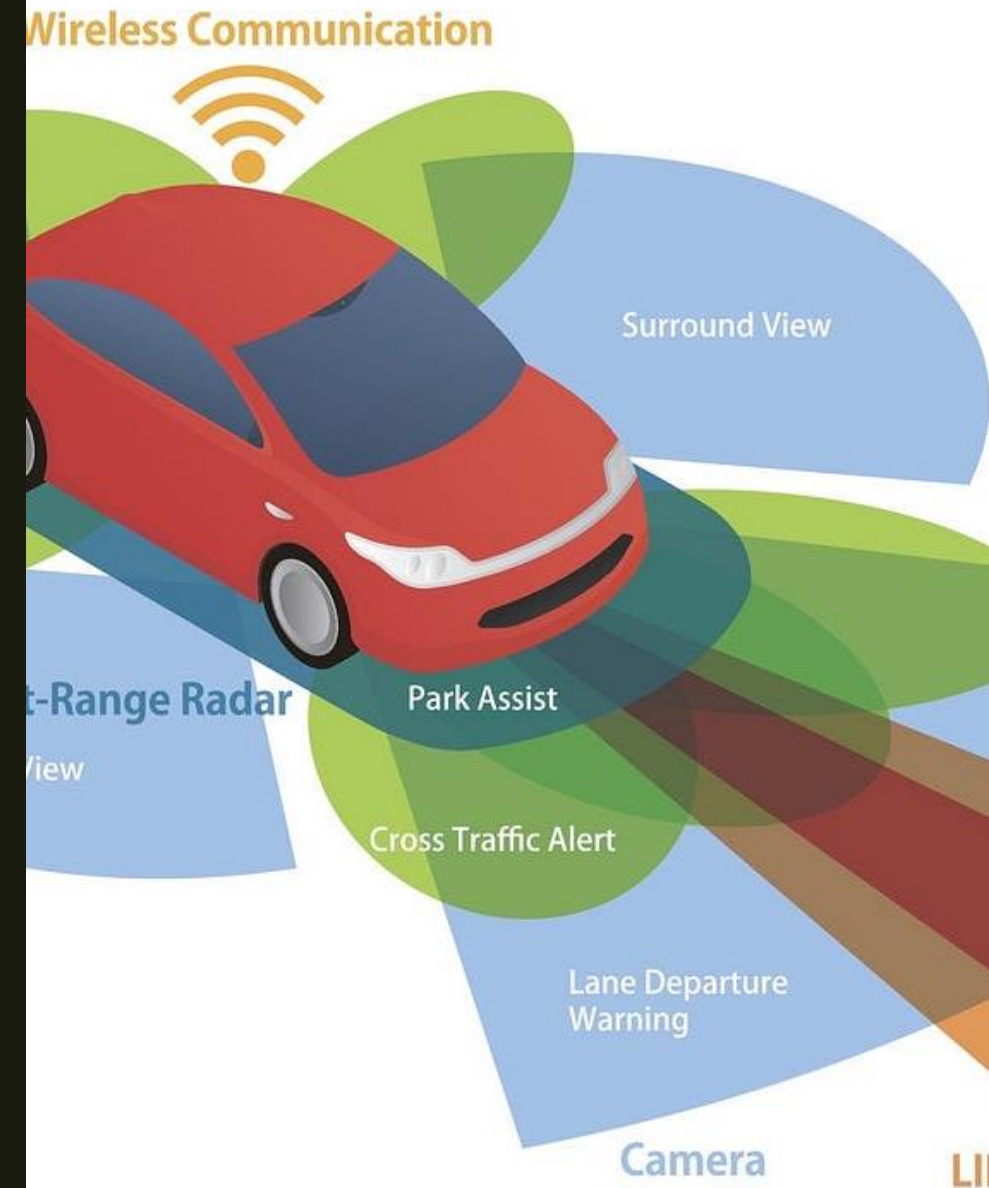
Contact and Force Sensors

- **Tactile Sensors:**
 - *Detect touch, pressure, or force.*
 - **Applications:** Grasping, manipulation, object detection.
- **Force/Torque Sensors:**
 - *Measure force and torque applied to an object.*
 - **Applications:** Robotic surgery, assembly tasks, haptic feedback.
- **Bump Sensors:**
 - *Detect physical contact with objects.*
 - **Applications:** Collision detection, obstacle avoidance.



Sensor Fusion

- The process of combining data from multiple sensors to obtain a more accurate and reliable perception of the environment.
- Techniques:
 - **Neural Networks:** Combines learned features from multiple inputs from sensors to classify its surrounding.
 - **Kalman Filter:** Combines predictions from a model with sensor data to estimate the state of a system.
 - **Particle Filter:** Uses a set of particles to represent the distribution of the state and updates them based on sensor measurements.



Perception Systems

- **SLAM (Simultaneous Localization and Mapping):**
 - *A method used to build a map of an environment while simultaneously tracking the robot's location within it.*
 - *Visual SLAM, LiDAR-based SLAM.*
 - **Applications:** Autonomous navigation, mapping unknown environments.
- **Object Recognition:**
 - *Identifying and classifying objects in the environment.*
 - *Deep learning, feature matching.*
 - **Applications:** Planning, Pick-and-place tasks, surveillance, human-robot interaction.



Challenges in Robot Perception

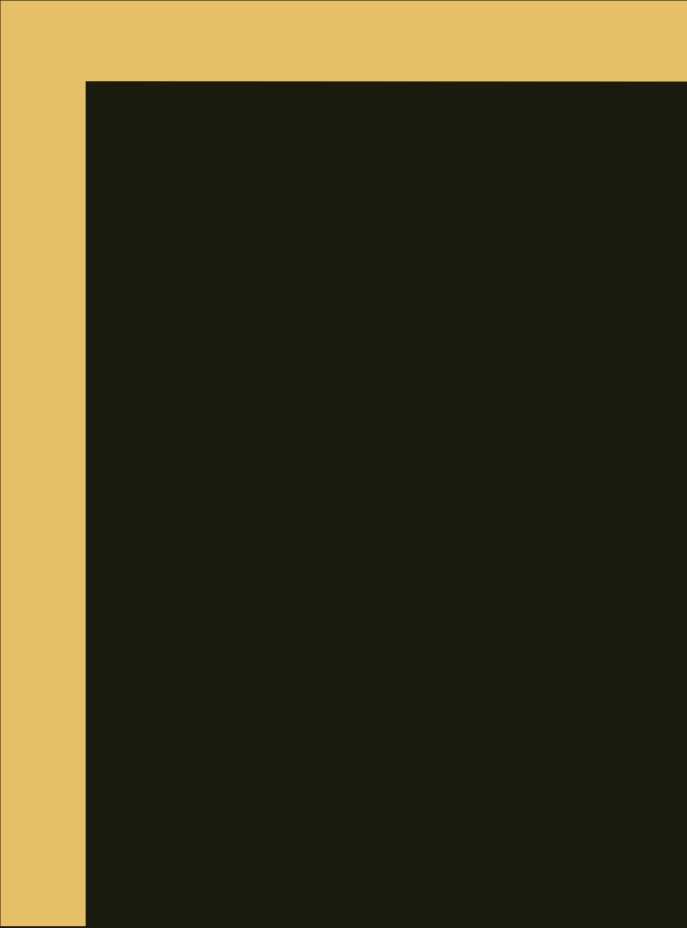
- **Data Quality and Reliability:** Ensuring sensors provide accurate and reliable data in various conditions.
- **Computational Complexity:** Processing and interpreting large amounts of sensory data in real-time.
- **Sensor Integration:** Combining data from different sensors seamlessly.
- **Environmental Factors:** Dealing with changes in lighting, weather, and terrain that affect sensor performance.



Summary

- **Types of Sensors:** Vision, proprioceptive, contact, force, and environmental sensors.
- **Perception Systems:** SLAM, object recognition, sensor fusion.
- **Applications:** Autonomous vehicles, industrial automation, healthcare, service robotics.





Q&A